GENERAL NOTES :

1. MINIMUM DESIGN REQUIREMENTS TO BE PER CAN/CSA S16.0 STANDARD LATEST EDITION, UNLESS OTHERWISE NOTED HEREIN
2. PAINT: AS PER SPECIFICATIONS PROVIDED.
3. THE ISSUANCE OF THIS DRAWING DOES NOT CONSTITUTE THE ACCEPTANCE OF A CUSTOMER'S ORDER
4. THE DESIGN IS BASED UPON LOAD INFORMATION SPECIFICALLY SUBMITTED NO SPECIAL LOADS OROTHER FORCES HAVE BEEN PROVIDED FOR UNLESS MENTIONED IN STRUCTURAL DRAWINGS SUCH SPECIAL LOADS OR OTHER FORCES SHALL INCLUDE, WITHOUT LIMITATION: UPLIFT,CONCENTRATED LOADS FROM ROOF TOP UNITS, AXIAL LOADS FROM KICKER ANGLES, ETC.
5. ERECTION DRAWINGS HEREIN WERE PREPARED USING THE STRUCTURAL PORTION OF THE CONTRACT DRAWINGS AS ITS PRIMARY GUIDE USING THE ARCHITECTURAL DRAWINGS (WHEN PROVIDED) ONLY FOR MISSING INFORMATION OR FOR CLARIFICATION. DOES NOT ACCEPT ANY RESPONSIBILITY FOR DISCREPANCIES BETWWEN THE STRUCTURAL AND ARCHITECTURAL DRAWINGS

CONTRACTOR NOTES :

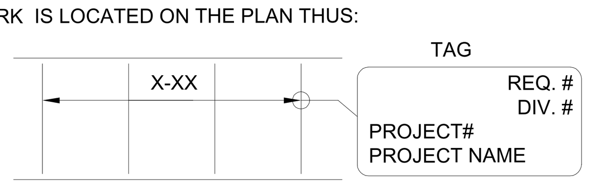
1. OWSJ TOP CHORDS AND SEAT WIDTHS MAY VARY.
2. FOR A COMPLETE PACKAGE OF STANDARDS & CONNECTION DETAILS PLEASE REFER JOIST DESIGN DOCUMENT.
3. CONFIRM ALL JOIST SHOE DEPTHS WITH CONNECTION DETAILS PRIOR TO DESIGN DOCUMENT.
4. REVIEW ENCLOSED LAYOUT FOR CONFORMANCE AND CONNECTION METHODS. INFORM OF ANY MISINTERPRETED CONNECTION DETAILS OR DIMENSIONAL DISCREPANICES TO ENSURE AS BUILT CONDITIONS ARE SATISFIED.
5. CONTRACTOR TO CONFIRM OR SUPPLY ALL CLOUDED INFORMATION WHERE SHOWN ON PLAN.

APPROVAL NOTES : (IF APPLICABLE)

1. PLEASE REVIEW THESE DRAWINGS CAREFULLY, IT REPRESENTS OUR INTERPRETATION OF THE CONTRACT DOCUMENTS.
2. INTERFERENCES WITH BRIDGING LINES SHALL BE SITE COORDINATED AND UNLESS OTHERWISE DETAILED BY THE JOIST FABRICATOR,TYPICAL "DETAIL HH" SHOULD BE FOLLOWED
3. SUBSEQUENT CHANGES TO INFORMATION SHOWN ON THESE DRAWINGS AFTER FIRST SUBMISSION COULD BE CONSIDERED AS CONTRACT CHANGES IF CONSIDERED A CONTRACT CHANGES THE ARCHITECT/ENGINEER SHOULD BE INFORMED IN WRITING.
4. UNLESS NOTED TO THE CONTRARY ON THESE DRAW.INGS WHEN RETURNED FROM APPROVAL, IT WILL BE CONSIDERED THAT THE INFORMATION SHOWN HEREIN HAS BEEN ACCEPTED BY ALL PARTIES.
5. APPROVER TO CONFIRM OR SUPPLY ALL CLOUDED INFORMATION WHERE SHOWN ON PLAN.

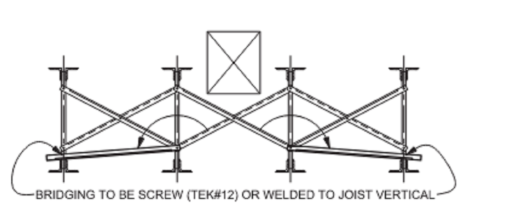
ERECTOR NOTES :

1. THIS DRAWINGS IS TO BE USED ONLY FOR THE ERECTION INDICATED BY AN ERECTION MARK ON THE PLANS AND/OR SECTIONS.
2. THE DESIGN AND MANUFACTURE OF THE MATERIALS ASSUMES THAT THEY ARE HANDLED IN ACCORDANCE WITH ALL APPLICABLE LAWS AND REGULATIONS. DESIGNERS IS NOT RESPONSIBLE FOR ANY MISHANDLING OR FAILURE TO PROPERLY ERECT THE MATERIALS
3. DESIGNERS HAS NOT EXAMINED ANY FIELD CONDITIONS AND ASSUMES NO RESPONSIBILITY FOR ANY SITE CONDITIONS. THE PURCHASER MUST NOTIFY OF ANY DISCREPANCIES BETWEEN THE FIELD CONDITIONS AND FILE AND FIELD USE DRAWINGS.
4. ANY MODIFICATION OF MATERIAL SUPPLIED WITHOUT PRIOR WRITTEN CONSENT WILL AUTOMATICALLY RELEASE DESIGNER FROM ALL LIABILITY WITH RESPECT TO SUCH MATERIAL.
5. OWSJ MUST BE PERMANENTLY FASTENED IN PLACE AND BRIDGING ATTACHED BY WELDING OR BOLTING AS SHOWN BEFORE ANY CONSTRUCTION LOADS ARE APPLIED. ALL FIELD WELDING TO BE CARRIED OUT BY WELDERS APPROVED BY C.W.B. TO THE REQUIREMENTS OF C.S.A. -W47 AND W59
6. BOLTED CONNECTIONS (BRIDGING AND JOISTS) SHALL BE "SNUG TIGHT". THE 2004 RCSC SPECIFICATION DEFINES IT AS A JOINT IN WHICH THE BOLTS HAVE BEEN INSTALLED IN ACCORDANCE WITH SECTIONS 8.1. NOTE THAT NO SPECIFIC LEVEL OF INSTALLED TENSION IS REQUIRED TO ACHIEVE THIS CONDITION, WHICH IS COMMONLY ATTAINED AFTER A FEW IMPACTS OF AN IMPACT WRENCH OR THE FULL EFFORT OF AN IRONWORKER WITH AN ORDINARY SPUD THE PILES ARE SOLIDLY SEATED AGAINST EACH OTHER. BUT NOT NECESSARILY IN CONTINUOUS CONTACT.
7. BACK CHARGES FOR REPAIRS OR ALTERATION TO PRODUCTS WILL NOT BE ACCEPTED UNLESS APPROVED PRIOR
8. JOISTS TO BE ERECTED SO THAT THE METAL TAG AT ONE END CORRESPONDS WITH THE END OF THE JOIST WHERE THE MARK IS LOCATED ON THE PLAN THUS:



BRIDGING INTERFERENCE/REMOVAL:

1. REMOVE BRIDGING FORM LOCATION OF DESIRED OPENING.
2. RELOCATE REMOVED BRIDGING MEMBERS AS SHOWN BELOW.
3. BOLT BRIDGING @JOIST BOTTOM CHORD NEAR OPENING AND FIX THE OTHER END TO ADJACENT JOIST AS PER PICTURE BELOW.



MAX. JOIST SPACING	MAX. HORIZ. BRIDGING SIZE
≤ 1720 / 5' - 7"	L1¼" x 1¼"
≤ 2240 / 7' - 4"	L1½" x 1½"
≤ 2620 / 8' - 7"	L1¾" x 1¾"
≤ 2970 / 9' - 9"	L2" x 2"

- IF SPACING IS GREATER THAN SHOWN
- SIMLAR CONCEPT TO BE APPLIED FOR WELDED BRIDGING

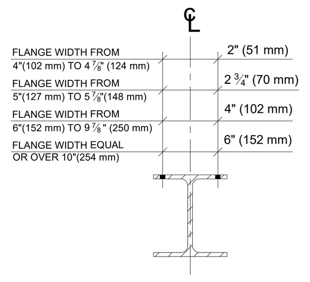
GENERAL JOIST/GIRDER SHOE INFORMATION:

NOTE: ALL ENCLOSED DETAILS AND INFORMATION MUST BE VERIFIED ON PLAN IF NOTED OTHERWISE. DETAILS ARE INTENDED AS STANDARD BUT MAY VARY WITH SPECIFIC CONTRACTOR, PROJECTS AND/OR CONDITIONS. ANY DISCREPANCIES BETWEEN GENERAL INFORMATION AND THE PLAN, THE PLAN WILL BE CONSIDERED AS CORRECT.

STANDARD GAUGES

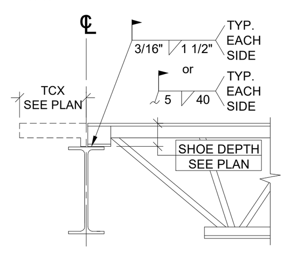
1. WHEN ONE JOIST BEARS AT JUNCTION OF TWO DIFFERENT BEAMS, THE SMALLER GAUGE IS TO BE USED
2. DESIGN CONDITIONS CAN DICTATE VARIANCES FOR THESE CONNECTIONS.
3. USE A325 3/4"∅ TYP. U.N.O.

FOR SIN BEAM & CONVENTION BEAM:



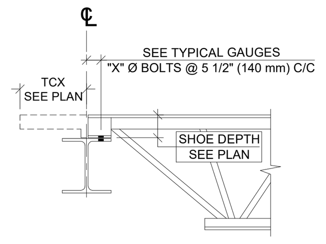
STANDARD WELDED SHOE

1. ONLY IF APPLICABLE - SEE PLAN
2. U.N.O. ON PLAN



STANDARD BOLTED SHOE

1. ONLY IF APPLICABLE - SEE PLAN
2. U.N.O. ON PLAN

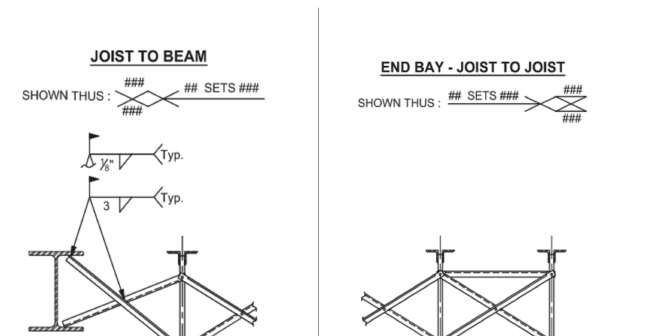


STANDARD BRIDGING DETAILS:

IMPORTANT NOTE: SEE PLAN FOR ADDITIONAL BRIDGING/BRACING DETAILS AND INSTRUCTIONS IF APPLICABLE.

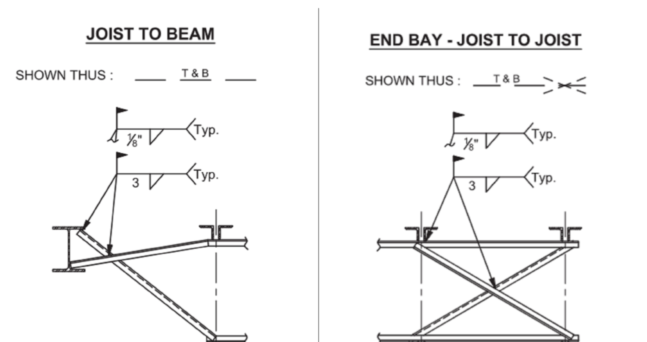
TYP. BOLTED BRIDGING

1. ONLY IF APPLICABLE - SEE PLAN
2. LOCATIONS OF X-BRIDGING NOT TO BE SCALED ON PLAN, EXACT LOCATIONS CAN BE REQUESTED IF REQUIRED.



TYP. WELDED HORIZONTAL BRIDGING

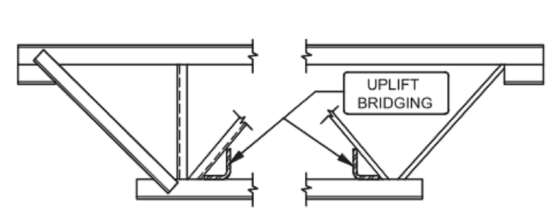
1. SEE PLAN FOR MATERIAL SIZES AND SPECIFICATIONS.
2. SEE PLAN FOR LOCATIONS OF WELDED HORIZONTAL BRIDGING, DO NOT SCALE THE PLAN.
3. HORIZONTAL BRIDGING CAN BE REQUIRED ON THE TOP AND/OR BOTTOM CHORDS. SEE PLAN SHOWN THUS:
4. BRIDGING SPLICE/LAP TO BE MAXIMUM 100mm (4")
5. QUANTITIES HORIZONTAL BRIDGING DETERMINED USING (O/A LENGTH (X) # ROWS)(+ 1/6%)



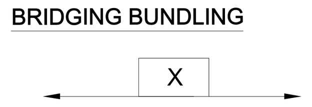
TYP. WELDED HORIZONTAL UPLIFT BRIDGING

1. SEE PLAN FOR MATERIAL SIZES AND SPECIFICATIONS.
2. SEE PLAN FOR LOCATIONS OF WELDED HORIZONTAL UPLIFT BRIDGING, LOCATED AT FIRST AND LAST PANEL POINTS TYP. U.N.O.
3. FOR INSTALLATION PROCEDURE SEE "TYP. WELDED HORIZONTAL BRIDGING" DETAILS ABOVE AND SUBTRACT TOP ROW EXCEPT AT JOIST TO JOIST END BAYS.

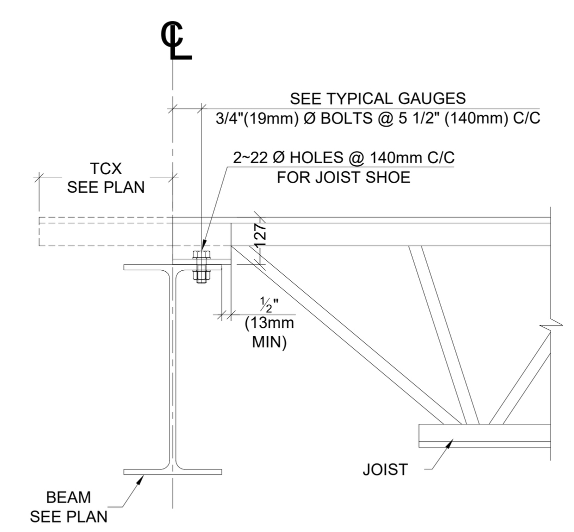
SHOWN THUS: __B___OR: __B__



BRIDGING BUNDLING

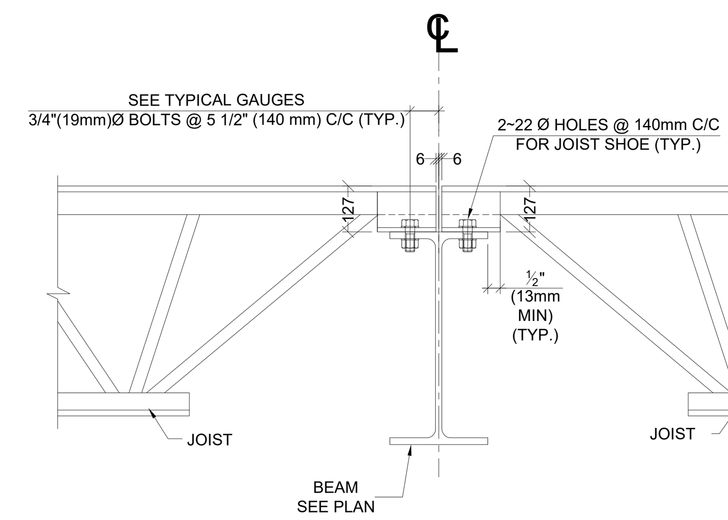


=REPRESENT BRIDGING BUNDLES



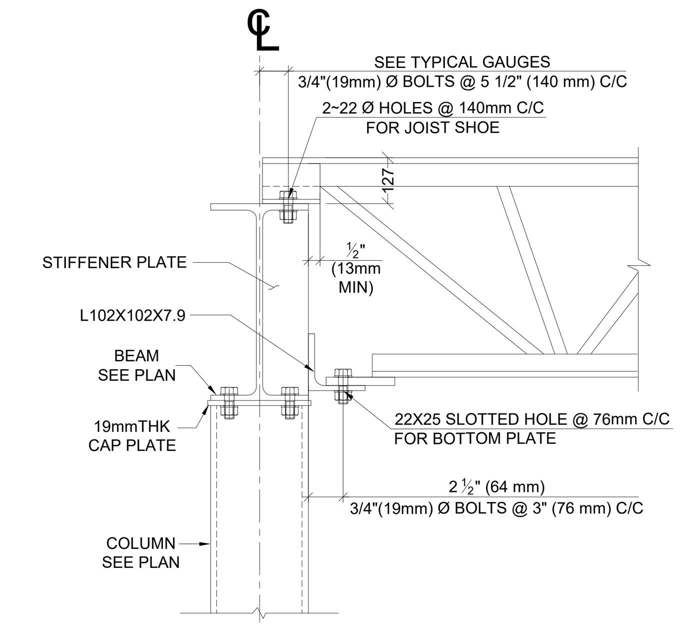
DETAIL - J1

TYPICAL BEAM TO JOIST CONNECTION DETAIL AT EXTERIOR LOCATIONS



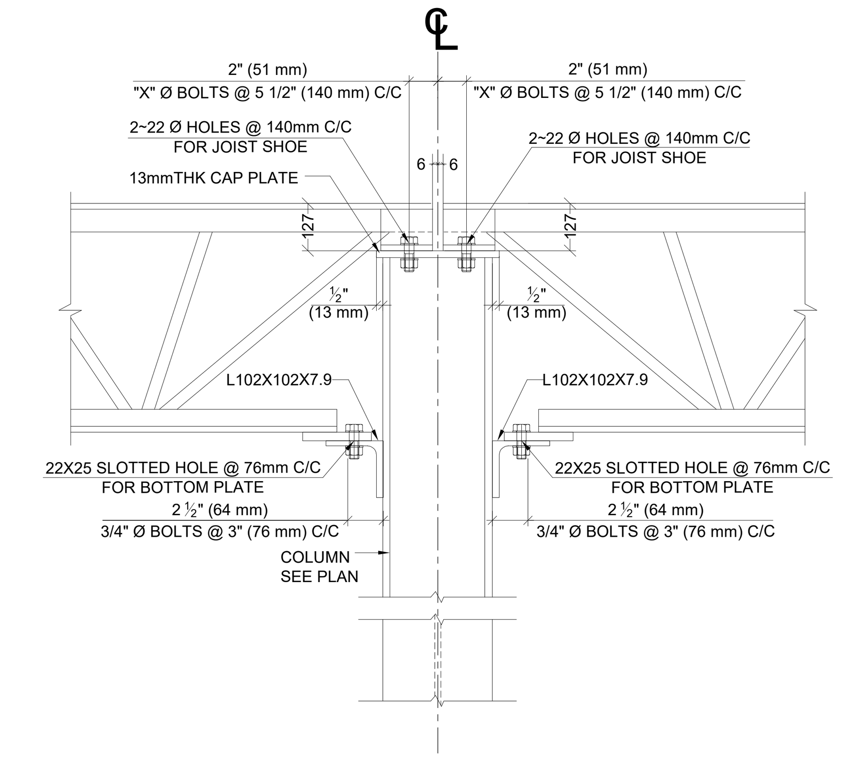
DETAIL - J2

TYPICAL BEAM TO JOIST CONNECTION DETAIL AT INTERIOR LOCATIONS



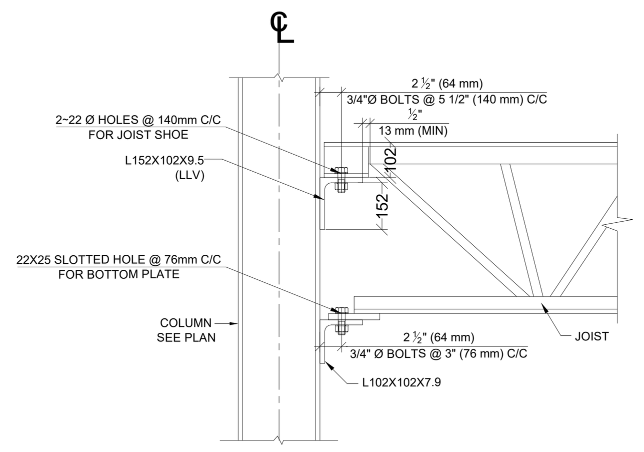
DETAIL - J3

TYPICAL COLUMN TO JOIST CONNECTION DETAIL AT INTERIOR COLUMN STIFFNER PLATE LOCATIONS



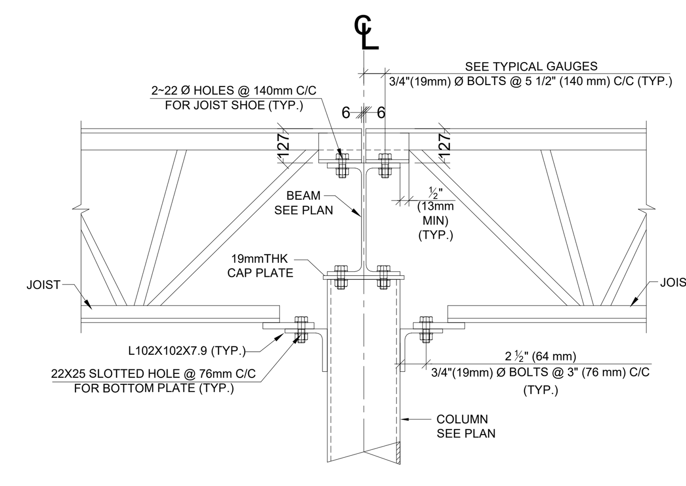
DETAIL - J4

TYPICAL CONNECTION DETAILS FOR W~COLUMN CAP PLATE TO JOIST AT INTERIOR COLUMN LOCATIONS



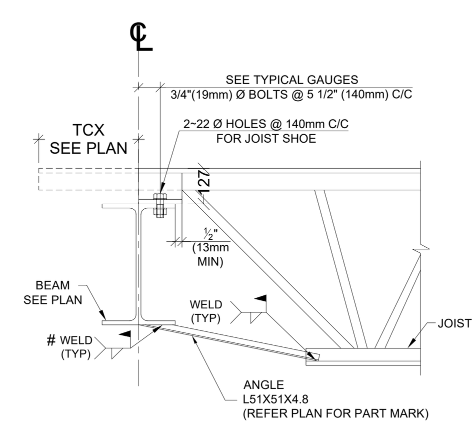
DETAIL - J5

TYPICAL SEAT CONNECTION DETAILS FOR JOIST CONNECTING TO COLUMN FACE



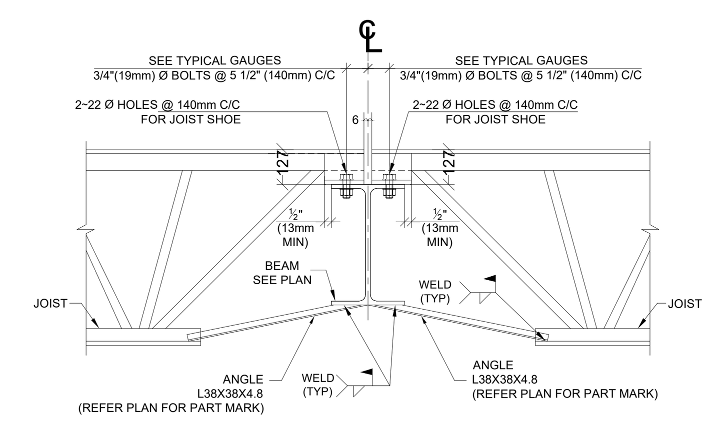
DETAIL - J6

TYPICAL COLUMN TO JOIST CONNECTION DETAIL AT INTERIOR COLUMN LOCATIONS



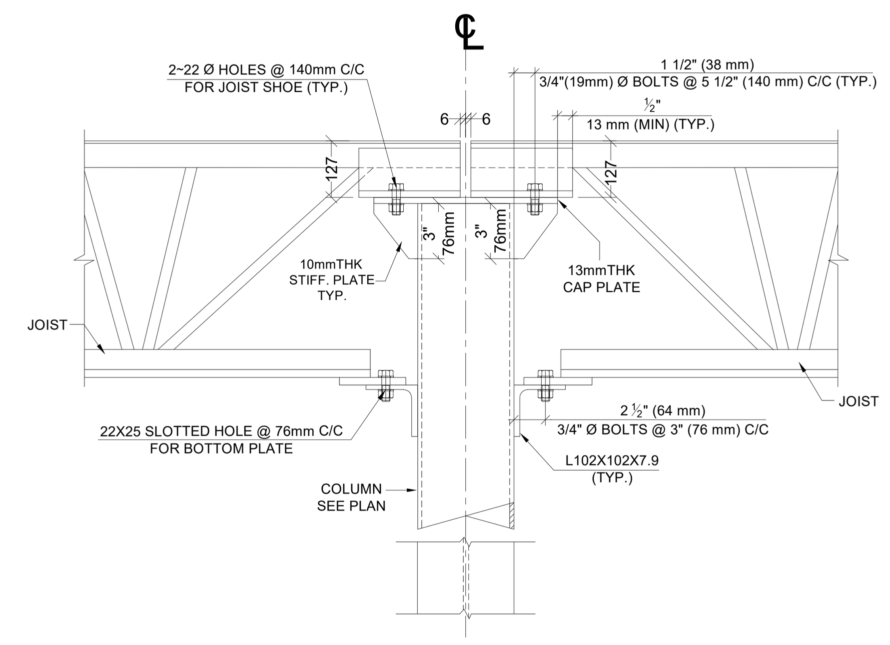
DETAIL - J7

TYPICAL BEAM TO JOIST CONNECTION DETAIL AT BOTTOM CHORD EXTENSION LOCATION (EXTENSION @ ONE SIDE)



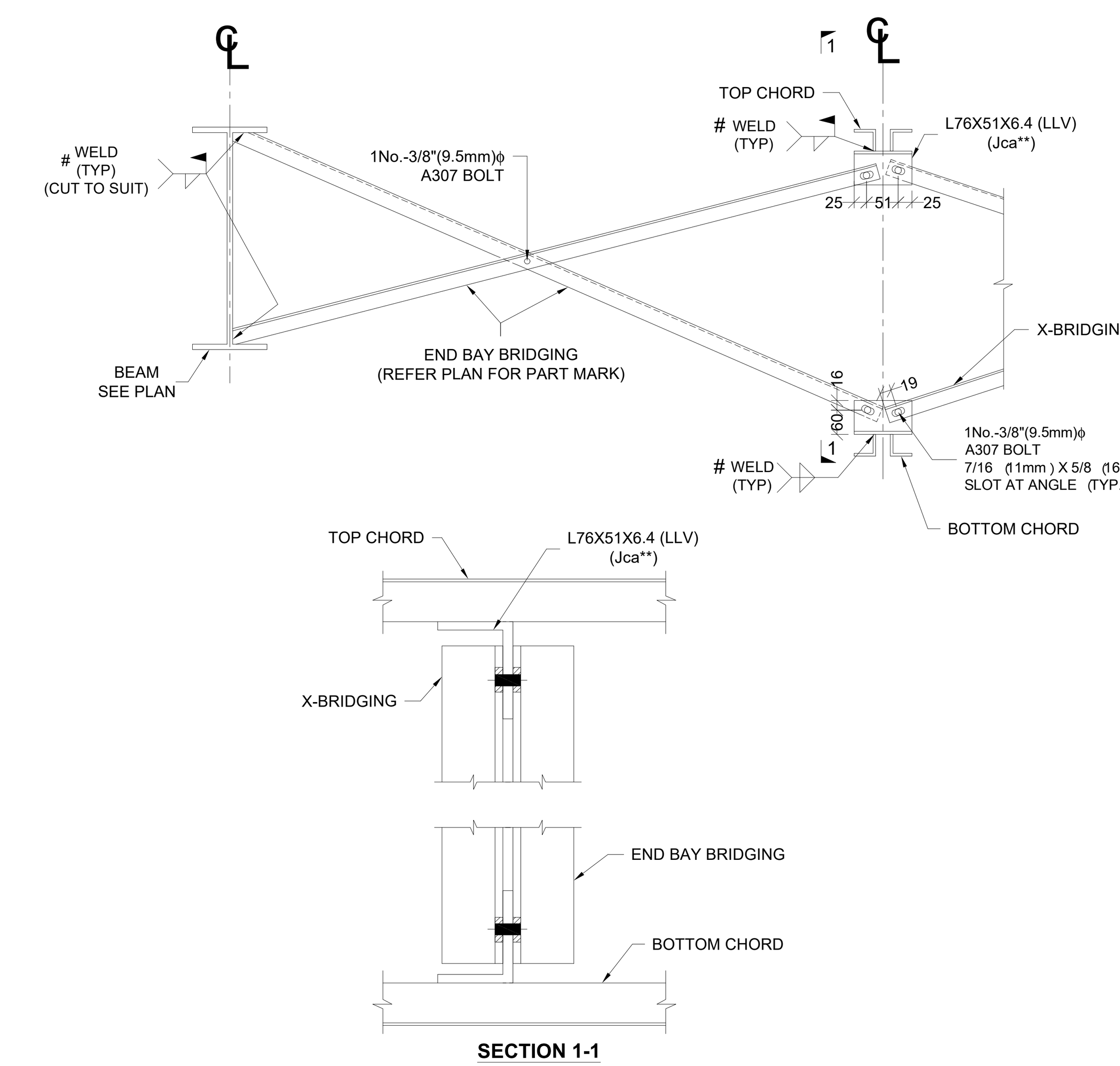
DETAIL - J8

TYPICAL BEAM TO JOIST CONNECTION DETAIL AT BOTTOM CHORD EXTENSION LOCATION (EXTENSION @ BOTH SIDE)



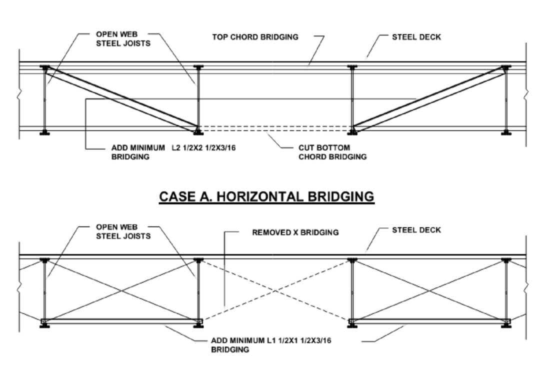
DETAIL - J9

TYPICAL CONNECTION DETAILS FOR HSS~COLUMN CAP PLATE TO JOIST AT INTERIOR COLUMN LOCATIONS



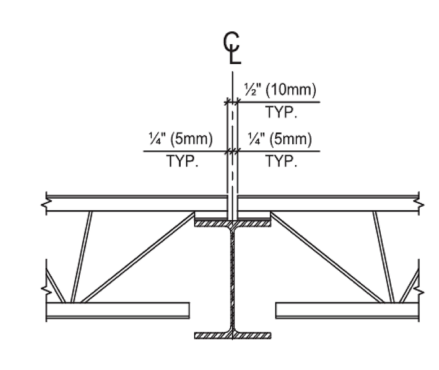
DETAIL - XX

TYPICAL BRIDGING CONNECTION DETAIL (UPLIFT BRIDGING & X-BRIDGING)



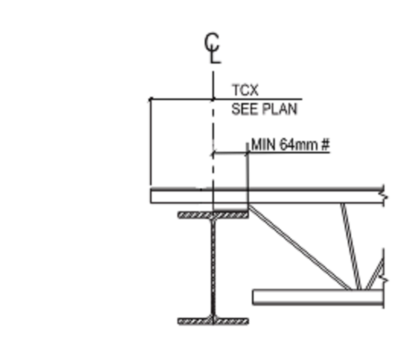
DETAIL - HH

TYP. DETAIL AT CUT JOIST BRIDGING



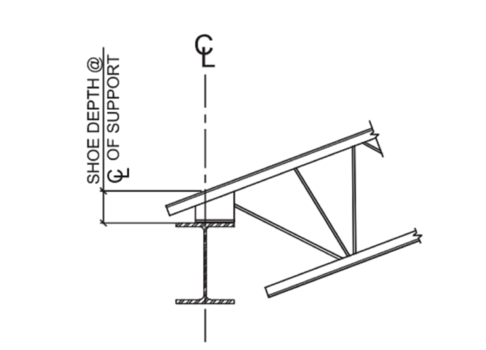
SHARED BEARING ON SUPPORTING BEAM:

1. EACH OWSJ REQUIRES A MIN.2½" (65mm) OF BEARING ON SUPPORT.



BEARING ON END BAY OR PERIMETER BEAM:

1. SHOE CONFIGURATION CAN VARY. VERIFY WITH TCX AS SHOWN ON PLAN.

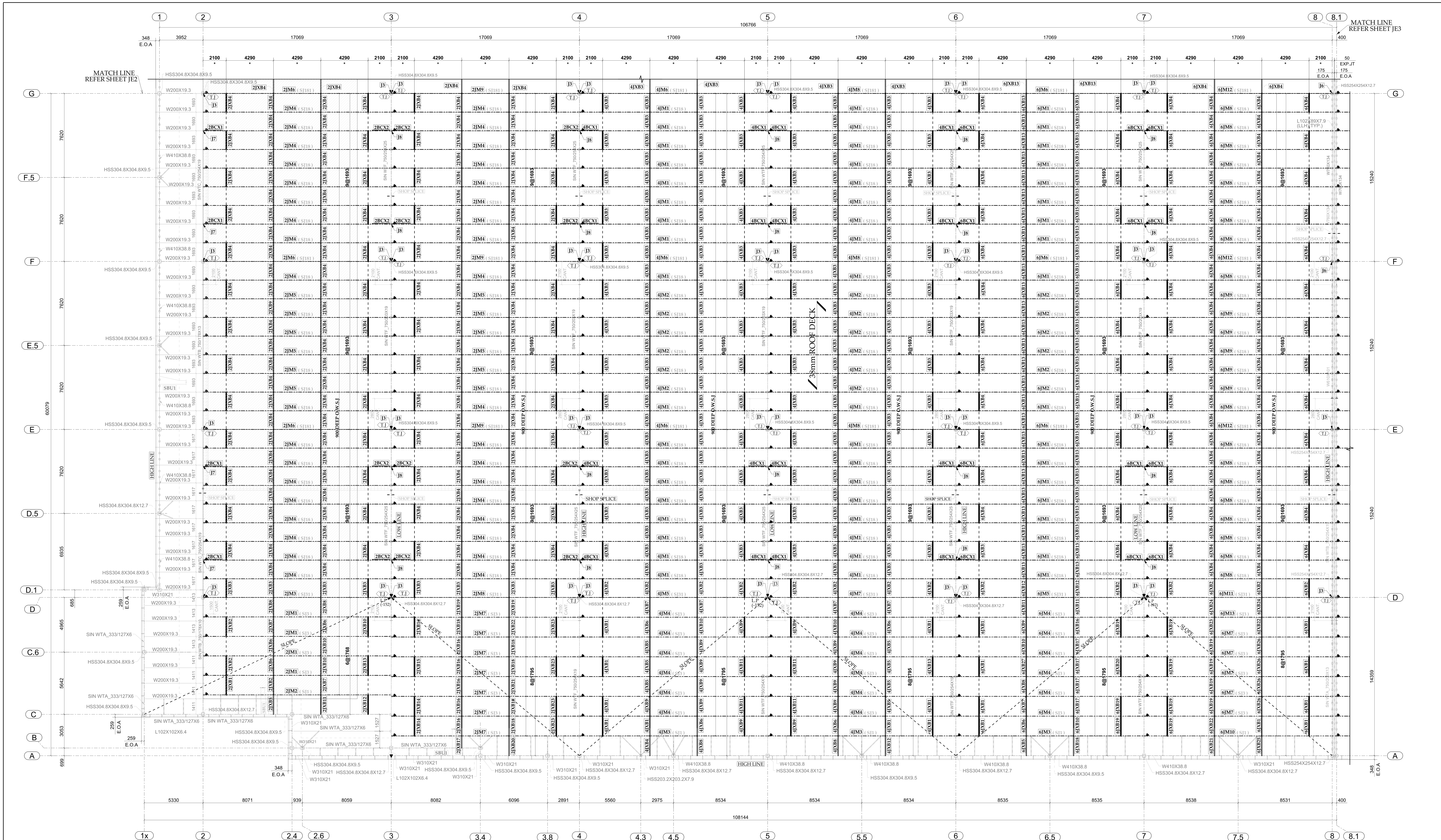


BEARING ON END BAY OR PERIMETER BEAM:

1. SHOE CONFIGURATION CAN VARY. VERIFY WITH TCX AS SHOWN ON PLAN.

LEGENDS

- LLV - LONG LEG VERTICAL
- BC - BOTTOM CHORD
- TCX - TOP CHORD EXTENSION
- BCX - BOTTOM CHORD EXTENSION
- LBCX - LOOSE BOTTOM CHORD EXTENSION
- T - BRIDGING @ TOP ONLY
- B - BRIDGING @ BOTTOM ONLY
- T & B - BRIDGING @ TOP & BOTTOM
- C/C - CENTER TO CENTER
- (±XX) - C HANGE IN ELEVATION
- EOD - EDGE OF DECK
- EOS - EDGE OF SHOE
- TOS - TOP OF STEEL
- FOW - FACE OF WALL
- F/S - FAR SIDE
- N/S - NEAR SIDE
- U/S - UNDER SIDE
- KB - KNEE BRACE
- O/A - OVERALL
- (S=X) - SHOE DEPTHS
- SIM - SIMILIAR
- TJ - TIE JOIST
- UNO - UNLESS NOTED OTHERWISE
- VIF - VERIFY IN FIELD
- (X) - JOIST CONNECTION TYPES



PARTIAL ROOF JOIST FRAMING PLAN

T/O STEEL = U/S OF DECK ELEVATION = 13327mm(U.N.O)

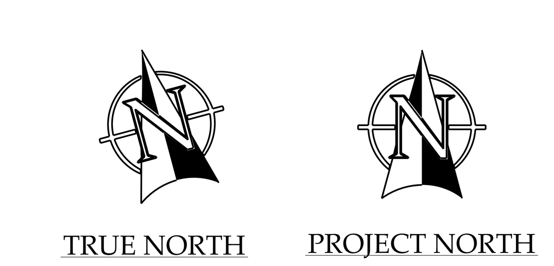
BLOCK-A

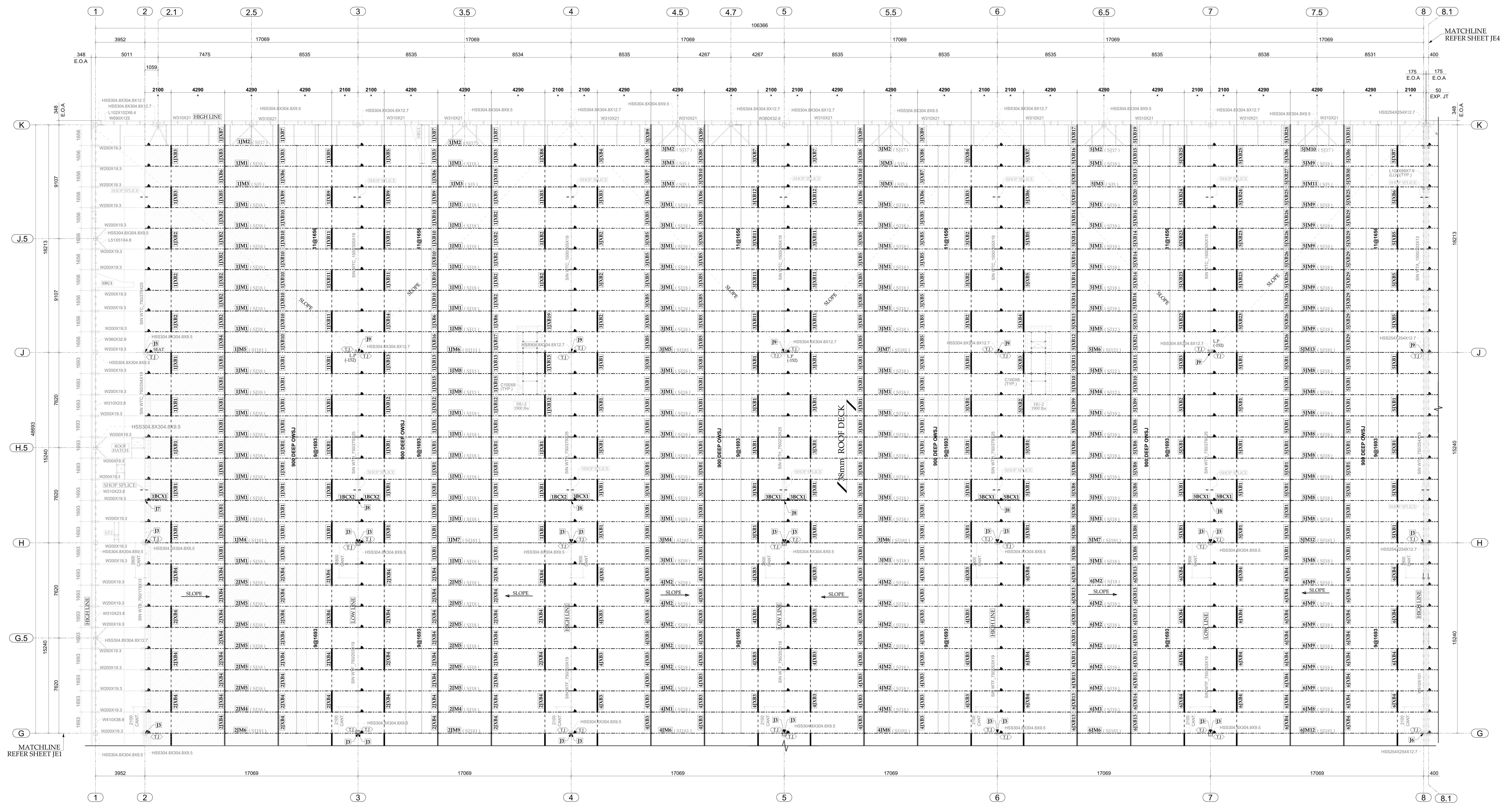
REFER SHEET JE4 FOR JOIST
SNOW DRIFT DIAGRAM & LOADING DETAILS

- **INDICATES 600mm WIDE TPO WALKWAY.**
- **INDICATES LOCATION OF SOLAR PANELS**

NOTES:

- TO STEEL = U/S OF ROOF DECK = EL. 13327mm.
- STANDARD SHOE DEPTH IS 5"(127mm) TYP.
- JOIST BEARING AT EXTERIOR BEAMS IS J1 TYP. U.N.O.
- JOIST BEARING AT INTERIOR BEAMS IS J2 TYP. U.N.O.
- TCX. - INDICATES TOP CHORD EXTENSION
- BCX. - INDICATES BOTTOM CHORD EXTENSION
- T.J. - INDICATES TIE JOIST
- TCX = EOA - 13mm (TYP.)
- DENOTES JOIST TAGGED END
- ★ - INDICATES JOIST BRIDGING LOCATING DIMENSION.





PARTIAL ROOF JOIST FRAMING PLAN

T/O STEEL= U/S OF DECK ELEVATION = 13327mm (U.N.O)

BLOCK - B

REFER SHEET JE4 FOR
SNOW DRIFT DIAGRAM & JOIST LOADING DETAILS

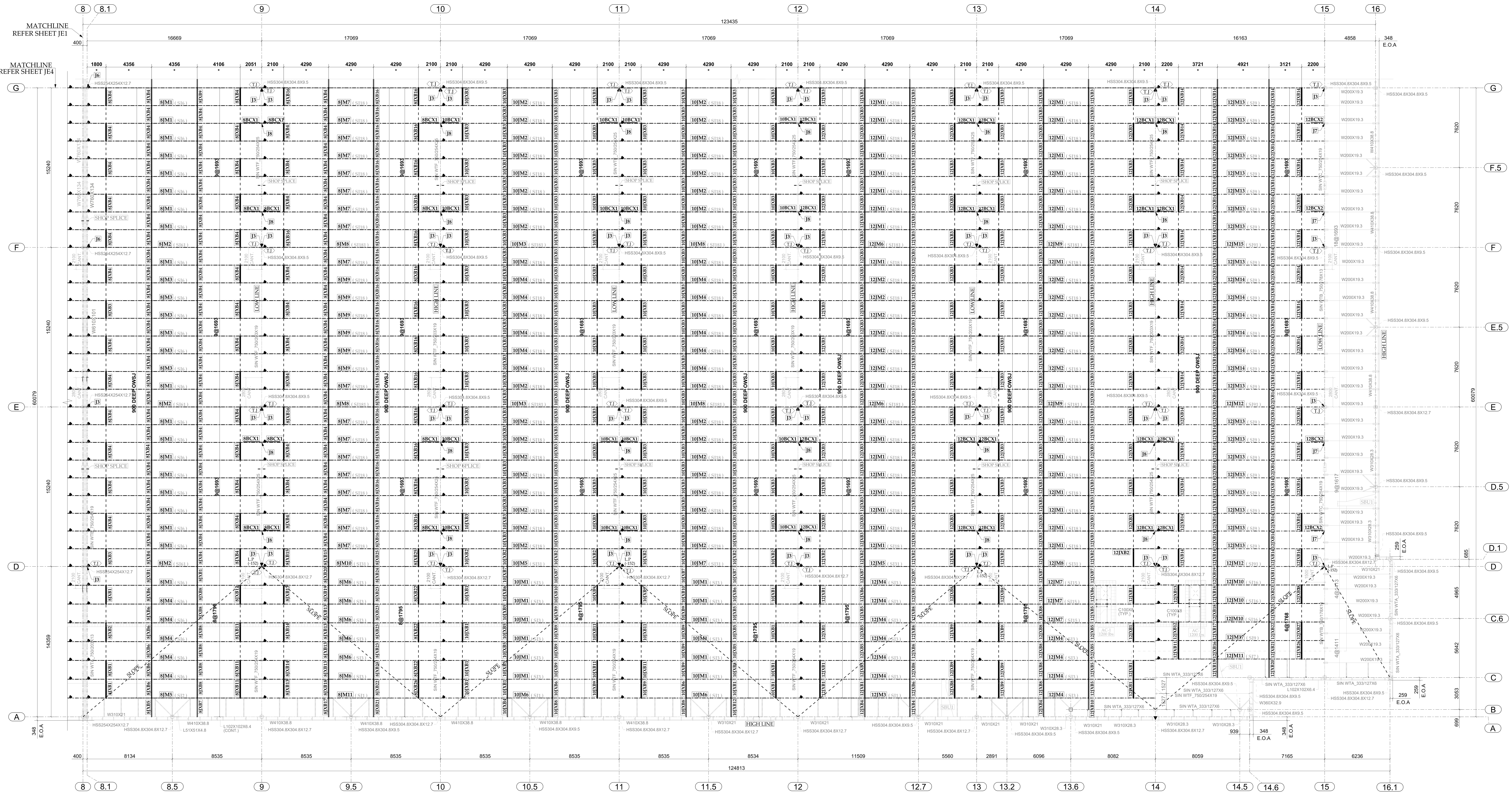
NOTES:

- TO STEEL = U/S OF ROOF DECK = EL. 13327mm.
- STANDARD SHOE DEPTH IS 5"(127mm) TYP.
- JOIST BEARING AT EXTERIOR BEAMS IS J1 TYP. U.N.O.
- JOIST BEARING AT INTERIOR BEAMS IS J2 TYP. U.N.O.
- TCX. - INDICATES TOP CHORD EXTENSION
- BCX. - INDICATES BOTTOM CHORD EXTENSION
- T.J. - INDICATES TIE JOIST
- TCX = EOA - 13mm (TYP.)
- DENOTES JOIST TAGGED END
- * - INDICATES JOIST BRIDGING LOCATING DIMENSION.

- INDICATES 600mm WIDE TPO WALKWAY
- INDICATES LOCATION OF SOLAR PANELS.

TRUE NORTH

PROJECT NORTH



PARTIAL ROOF JOIST FRAMING PLAN

T/O STEEL= U/S OF DECK ELEVATION = 13327mm (U.N.O)

BLOCK - C

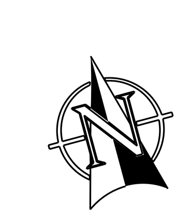
REFER SHEET JE4 FOR SNOW DRIFT DIAGRAM AND JOIST LOADING DETAILS

REFER SHEET JE5 FOR JOIST CONNECTION DETAILS

- 600mm WIDE TPO WALKWAY

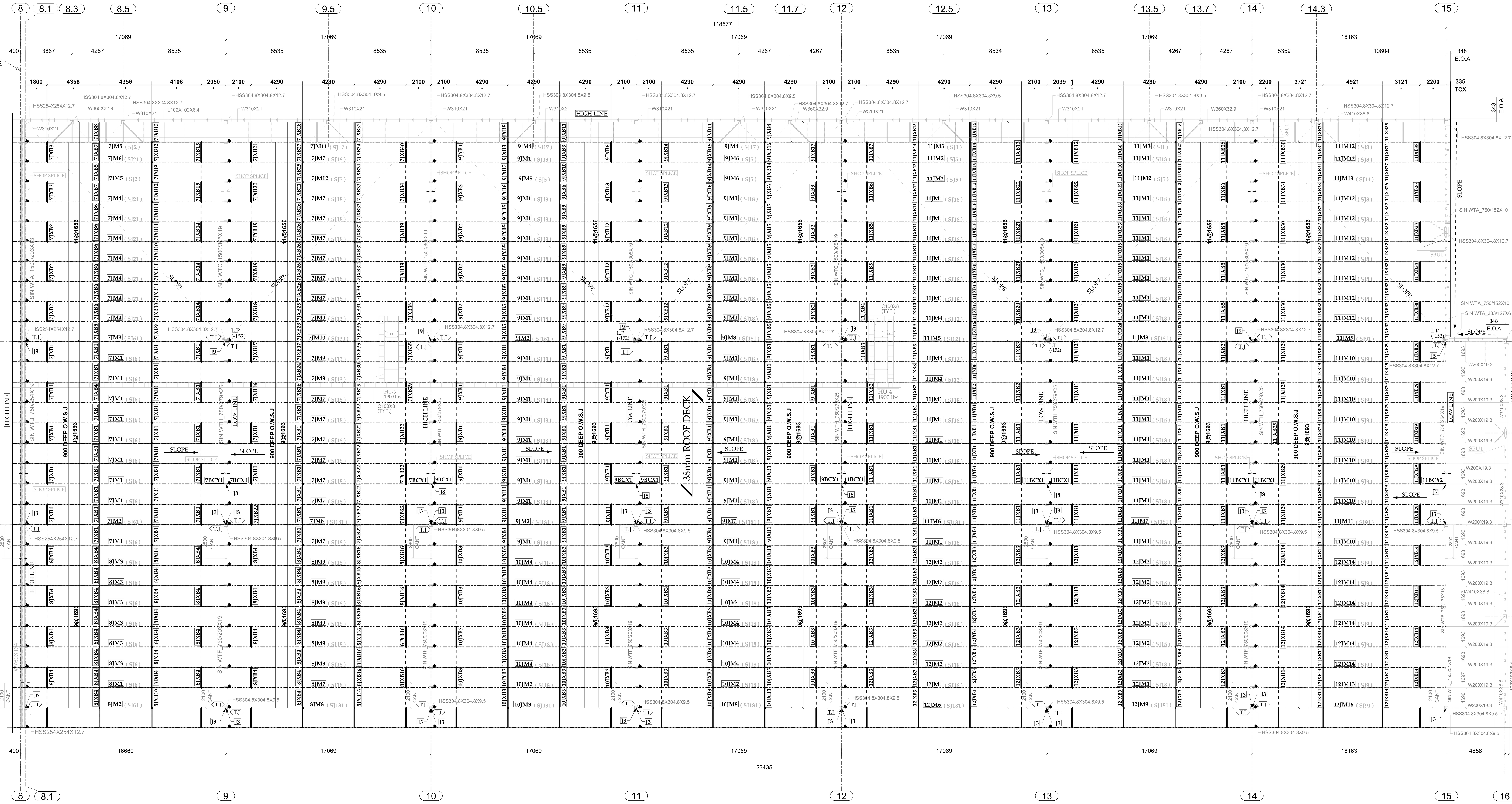
NOTES:

- TO STEEL = U/S OF ROOF DECK = EL. 13327mm.
- STANDARD SHOE DEPTH IS 5"(127mm) TYP.
- JOIST BEARING AT EXTERIOR BEAMS IS J1 TYP. U.N.O.
- JOIST BEARING AT INTERIOR BEAMS IS J2 TYP. U.N.O.
- TCX. - INDICATES TOP CHORD EXTENSION
- BCX. - INDICATES BOTTOM CHORD EXTENSION
- T.J. - INDICATES TIE JOIST
- TCX = EOA - 13mm (TYP.)
- DENOTES JOIST TAGGED END
- INDICATES JOIST BRIDGING LOCATING DIMENSION.



TRUE NORTH

PROJECT NORTH



PARTIAL ROOF JOIST FRAMING PLAN
(T/O STEEL = U/S OF DECK = 13327mm U.N.O)

BLOCK-D

REFER SHEET JE4 FOR SNOW DRIFT DIAGRAM AND JOIST LOADING DETAILS

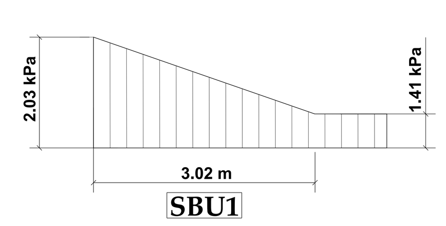
REFER SHEET JE5 FOR JOIST CONNECTION DETAILS

- 600mm WIDE TPO WALKWAY

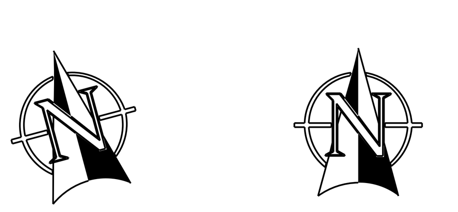
ROOF LOADING DETAILS	
ROOF LIVE LOADS	
GREATER EFFECT OF :	
SNOW 1.0[1.1(0.92x1.0x1.0x1.0)+0.4]	1.41 kPa
SNOW ACCUMILATION AT OBSTRUCTION FOR AREAS INDICATED ON PLAN	SEE PLAN
RAINWATER RETENTION	1.5 kPa
ROOF DEAD LOADS	
TPO ROOF	0.10 kPa
38mm METAL DECK:	0.10 kPa
OPEN WEB STEEL JOISTS (OWSJ):	0.20 kPa
STRUCTURAL STEEL:	0.20 kPa
MECHANICAL & ELECTRICAL:	0.25 kPa
SOLAR PV PANELS:	0.25 kPa
ROOF TOTAL DEAD LOAD	**1.10 kPa**
SUPER IMPOSED DEAD LOAD ON JOISTS DUE TO CONC. PAVERS = 0.7 kN/m (47 lbs/ft)	

NOTES:

- TO STEEL = U/S OF ROOF DECK = EL. 13327mm.
- STANDARD SHOE DEPTH IS 5"(127mm) TYP.
- JOIST BEARING AT EXTERIOR BEAMS IS J1 TYP. U.N.O.
- JOIST BEARING AT INTERIOR BEAMS IS J2 TYP. U.N.O.
- TCX. - INDICATES TOP CHORD EXTENSION
- BCX. - INDICATES BOTTOM CHORD EXTENSION
- T.J. - INDICATES TIE JOIST
- TCX = EOA - 13mm (TYP.)
- DENOTES JOIST TAGGED END
- ★ - INDICATES JOIST BRIDGING LOCATING DIMENSION.



SNOW BUILD UP (SBU) DIAGRAM



TRUE NORTH

PROJECT NORTH